Centre for
Data Analytics

Insight

A Semantic Web Platform for Automating the Interpretation of Finite Element Bio-simulations

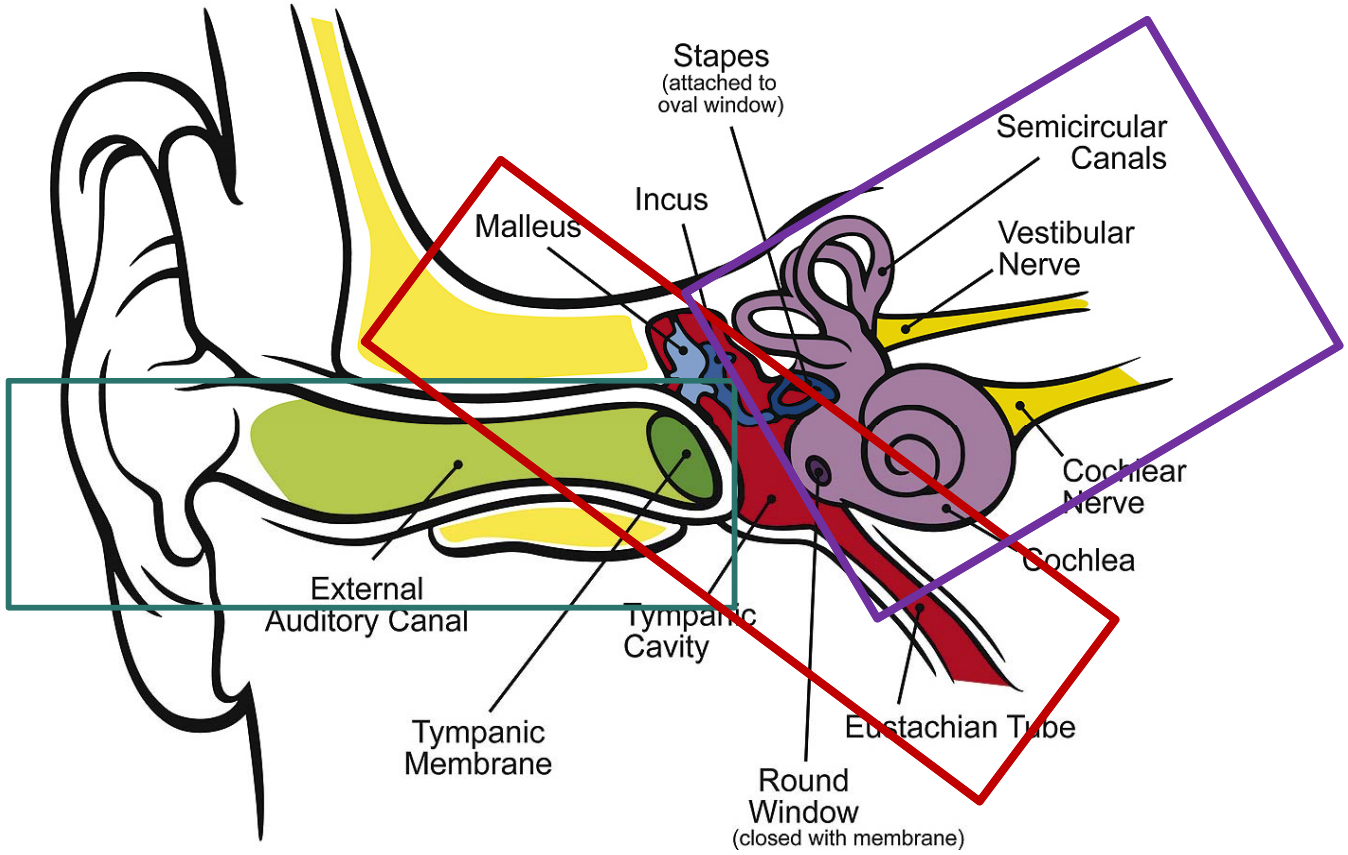
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10-12-2014

SWAT4LS-2014, Berlin
Germany

DCU NUI Galway OÉ Gaillimh UCC University College Cork, Ireland Coláiste na hOllscoile Corcaigh UCD DUBLIN SFI Science Foundation Ireland

Background – Hearing Loss

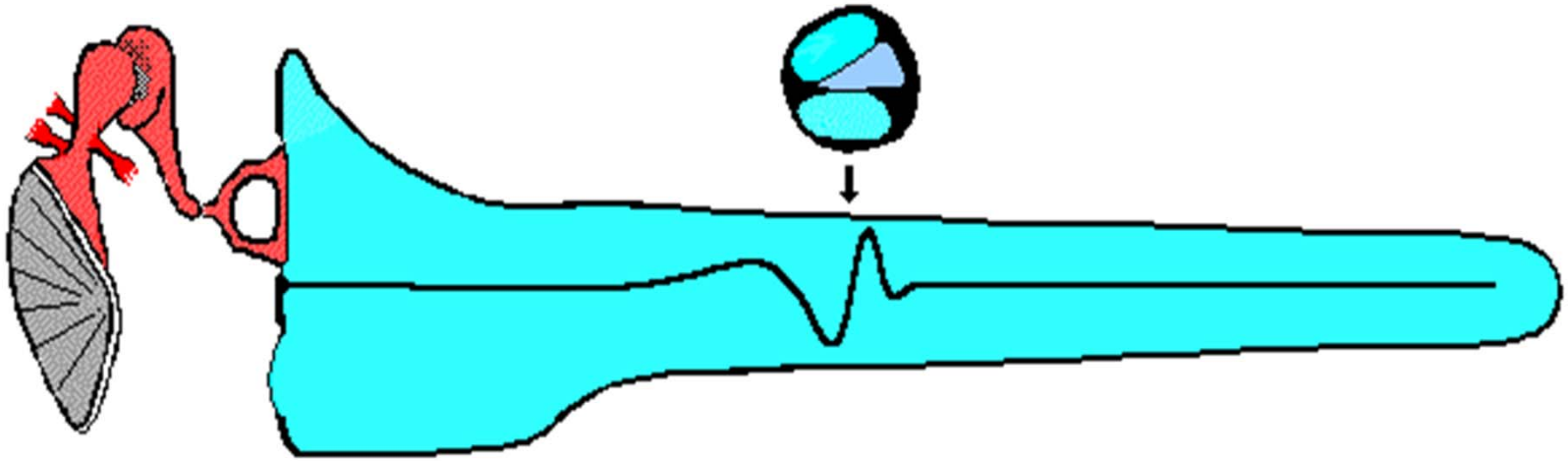


278 Million People

World Health Organization

- Outer ear gets excited both by the **sound waves** propagate through the ear canal and **strike the eardrum**
- In the middle ear the **ear drum vibrates** generating **pressure waves** in the inner ear **fluid chambers**
- The inner ear turns **pressure waves** into **electrical signals** that our **brain can understand**

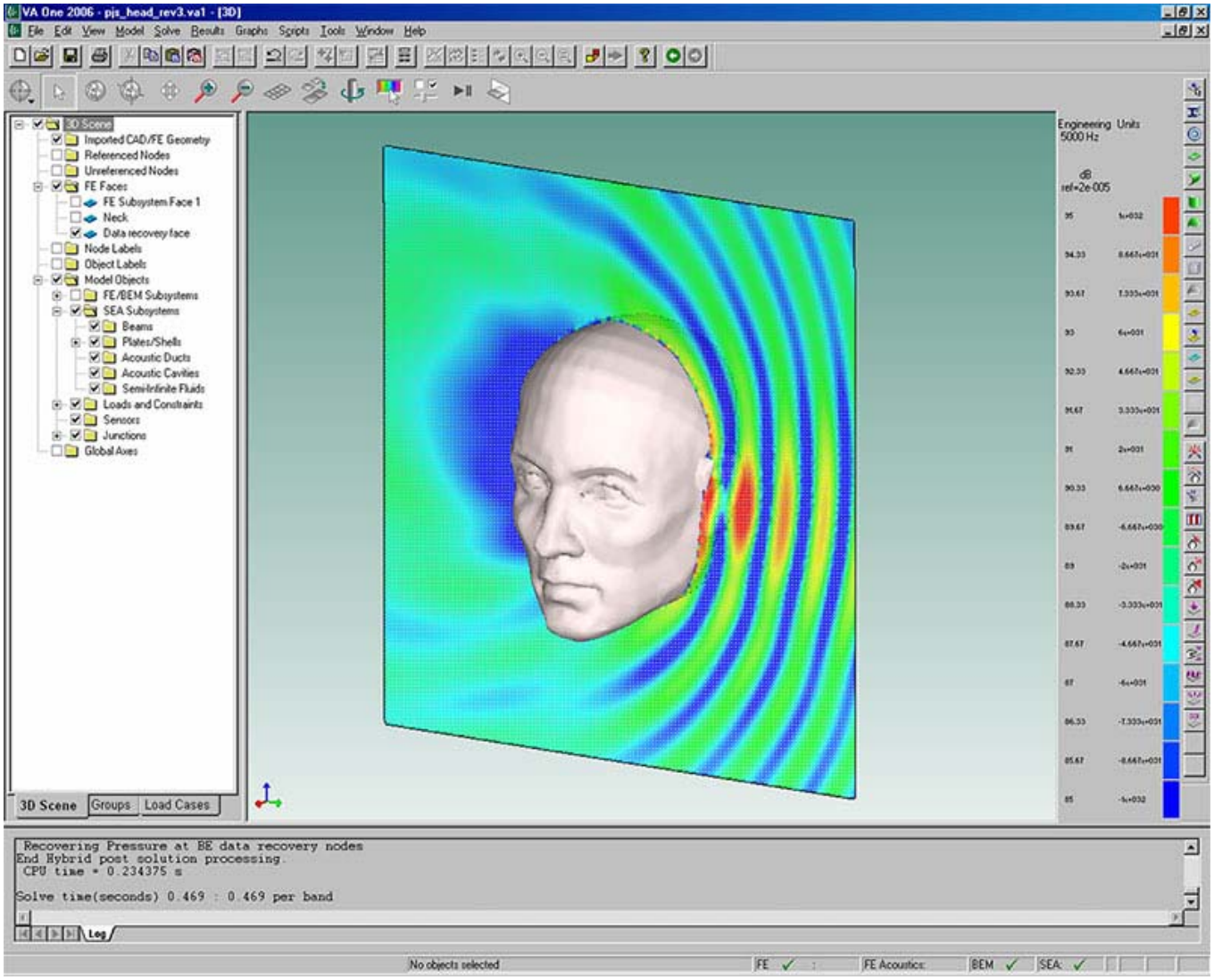
Background – Hearing Loss



- The **ear drum vibrates** generating **pressure waves** in the inner ear **fluid chambers**
- The inner ear turns **pressure waves** into **electrical signals** that our **brain can understand**

Infrastructure to integrate clinical knowledge, experimental data and inner ear models

Inner Ear - Bio Simulation Model & System



OpenFOAM

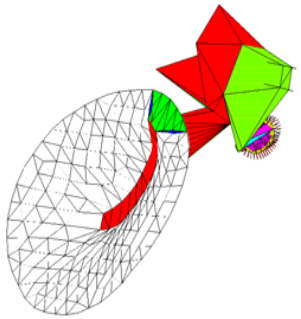
ANSYS

itk

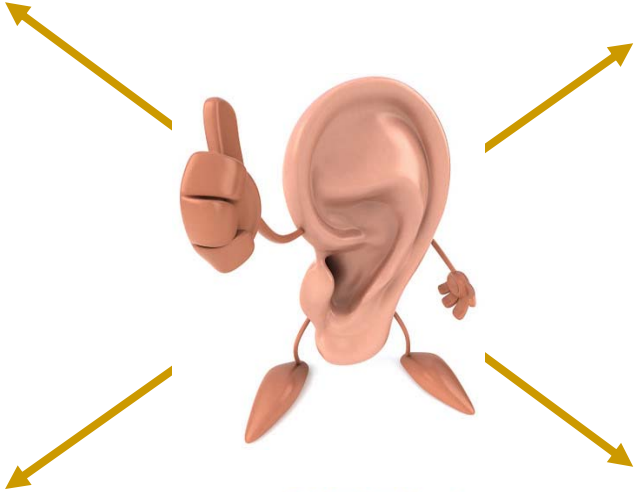
OpenCV

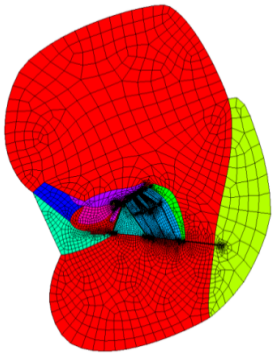
vtk

SIFEM Project

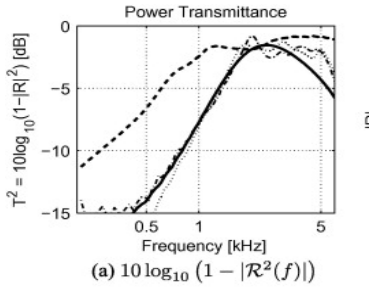


Finite Element Model



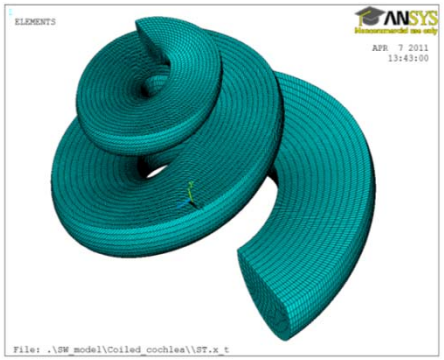


Micromechanics Model



Electrical Coupling Model

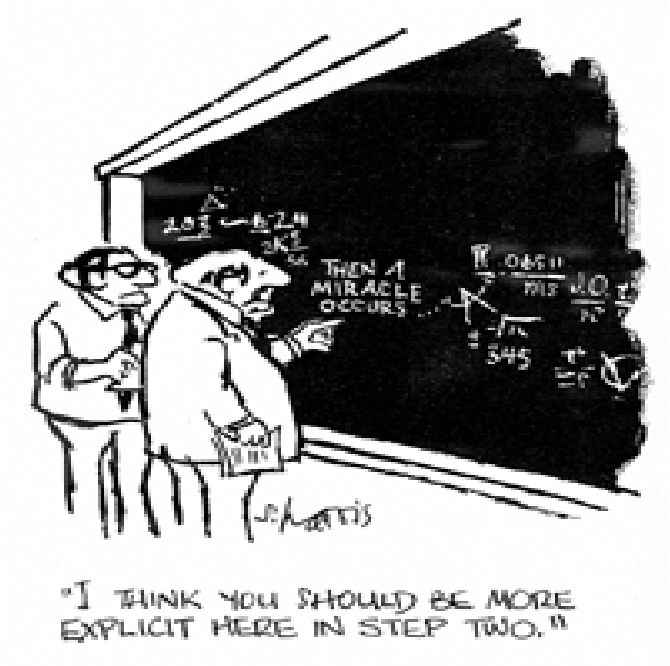
http://www.sifem-project.eu/



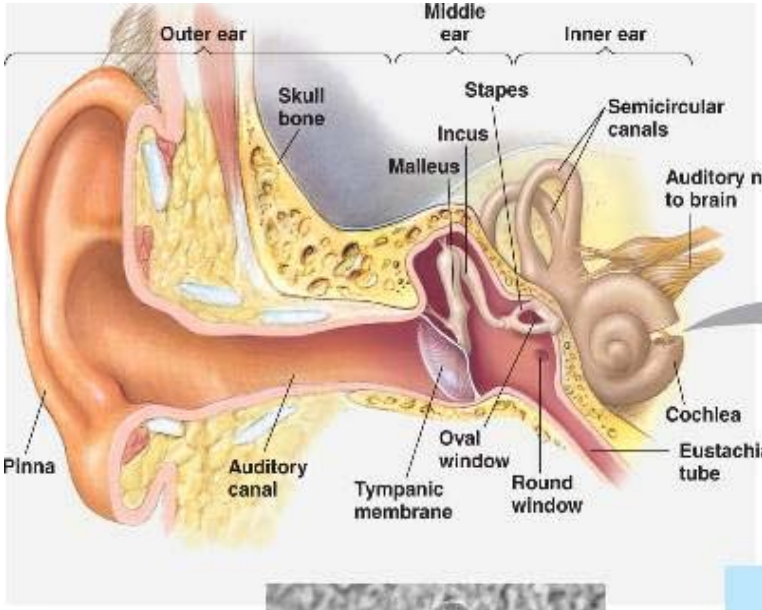
Fluid Coupling Model

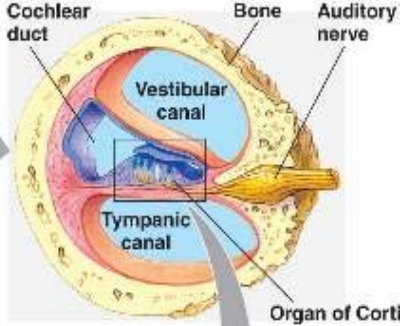
Goals

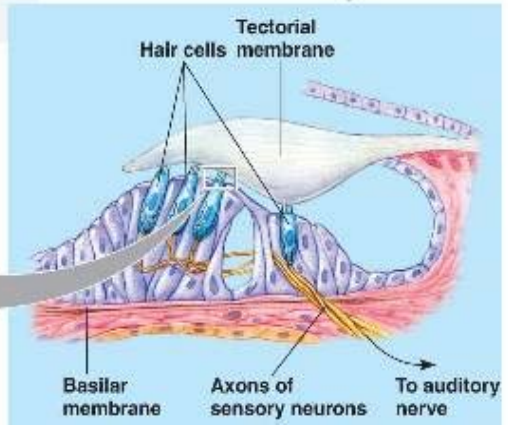
Automate the interpretation of finite element (FE) biosimulations ...

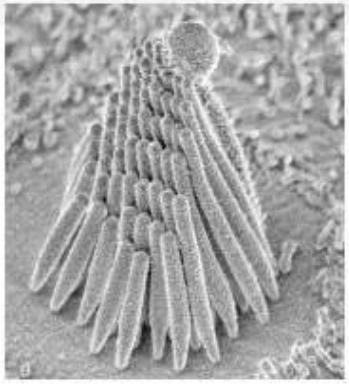


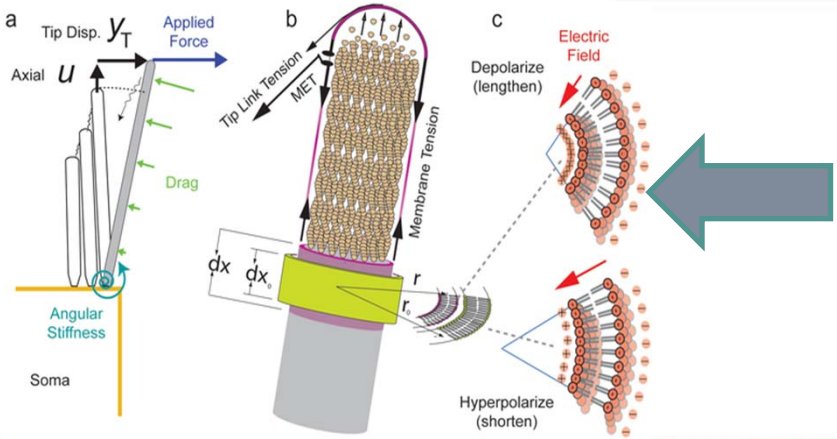
Motivational Scenario: Cochlear mechanics





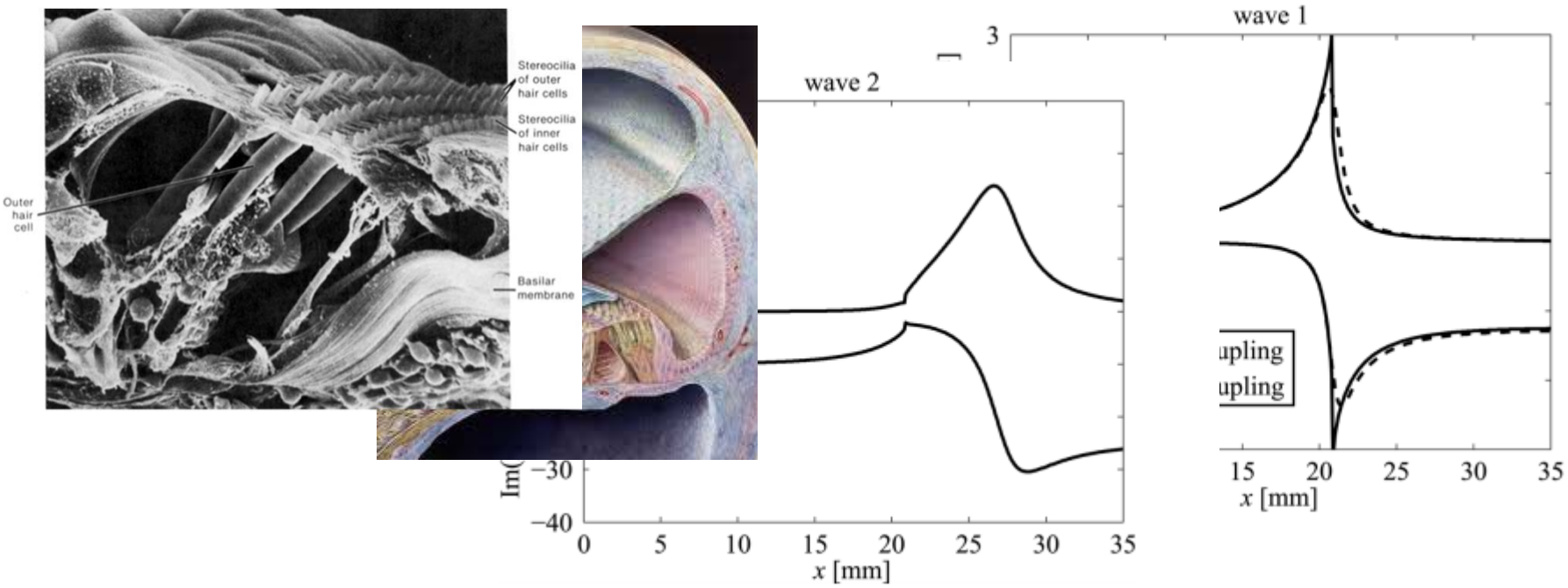






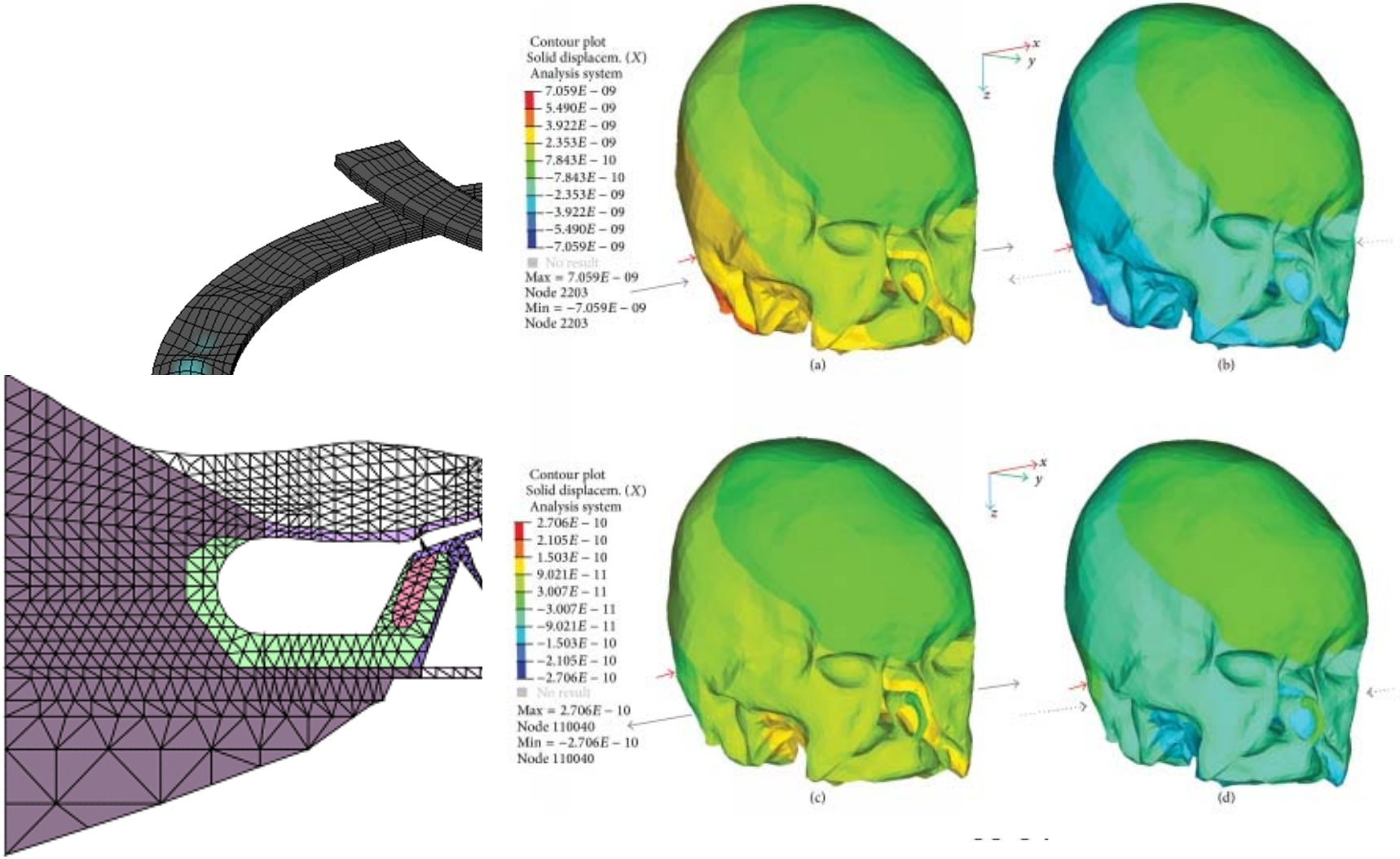
Characteristics of the FE Domain

- **Difficult to represent**
  - Physics, geometrical models, topological relations, algoithmic, mathematics



Dimensions of a FE Bio-simulation

Geometrical Model



Physics Model

- **FE equilibrium for solid**

$$\mathbf{M}\ddot{\mathbf{U}} + \mathbf{B}^{w}\dot{\mathbf{U}} + \mathbf{K}\mathbf{U} = \mathbf{F}_{nodal}$$

- **FE equilibrium for fluid**

$$\mathbf{M}\dot{\mathbf{V}} + \mathbf{K}_{w}\mathbf{V} + \mathbf{K}_{vp}\mathbf{P} = \mathbf{F}_{v}$$

Numerical Models/Solvers

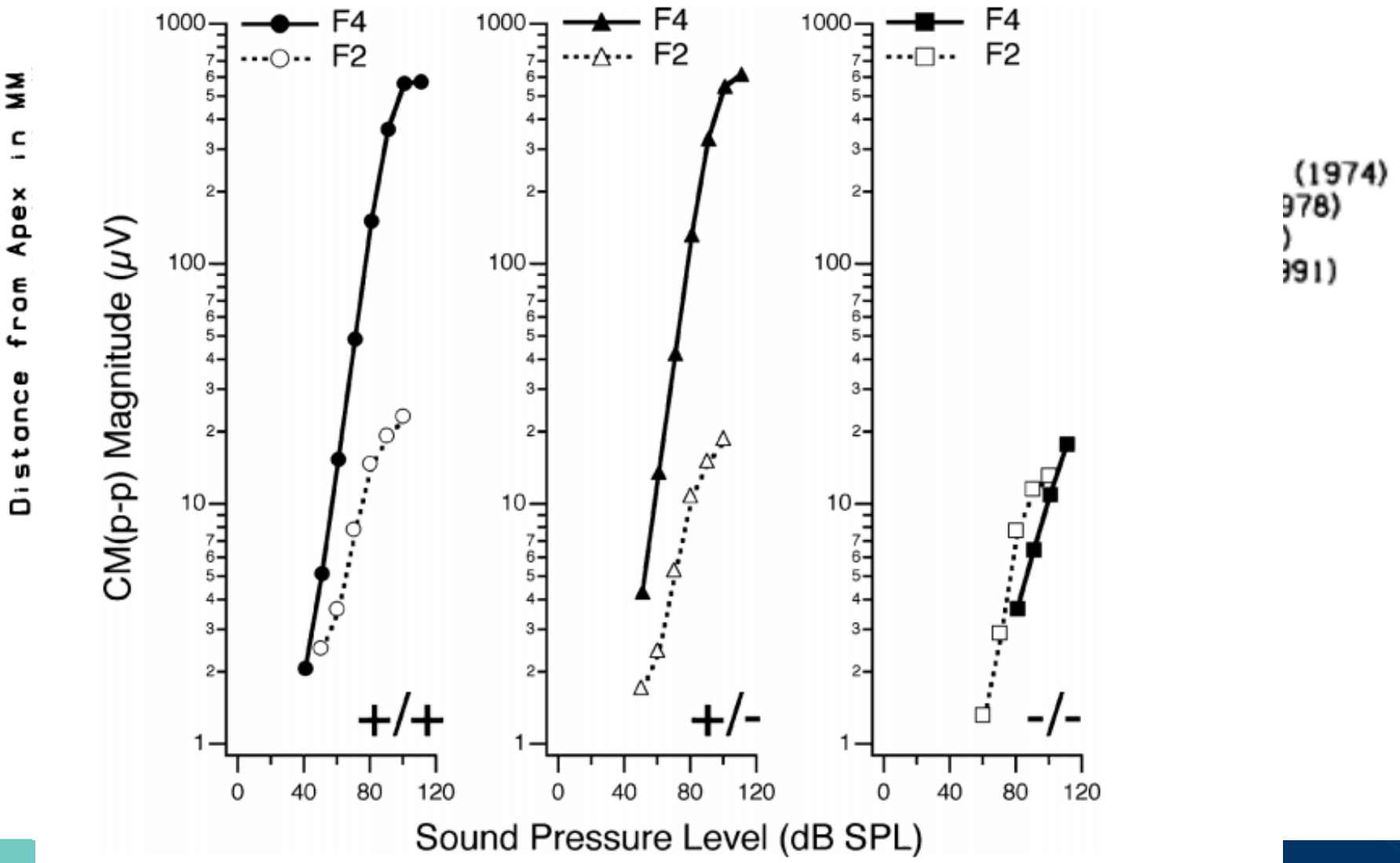
- Incremental-iterative implicit solution scheme

$$\begin{bmatrix} \frac{1}{\Delta t}\mathbf{M} + {}^{n+1}\tilde{\mathbf{K}}_{vv}{}^{(i-1)} & \mathbf{K}_{vp} \\ \mathbf{K}_{vp}^{T} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \Delta\mathbf{V}^{(i)} \\ \Delta\mathbf{P}^{(i)} \end{Bmatrix} =$$

$$\begin{Bmatrix} {}^{n+1}\mathbf{F}_{ext}^{(i-1)} \\ 0 \end{Bmatrix} - \begin{bmatrix} \frac{1}{\Delta t}\mathbf{M} + {}^{n+1}\mathbf{K}_{vv}{}^{(i-1)} & \mathbf{K}_{vp} \\ \mathbf{K}_{vp}^{T} & 0 \end{bmatrix} \begin{Bmatrix} {}^{n+1}\mathbf{V}^{(i-1)} \\ {}^{n+1}\mathbf{P}^{(i-1)} \end{Bmatrix} + \begin{Bmatrix} \frac{1}{\Delta t}\mathbf{M}\,{}^{n}\mathbf{V} \\ 0 \end{Bmatrix}$$

$$u_i = \sum_{K=1}^{N} N_K \quad r,s,t \quad U_i^K$$

Experimental Data



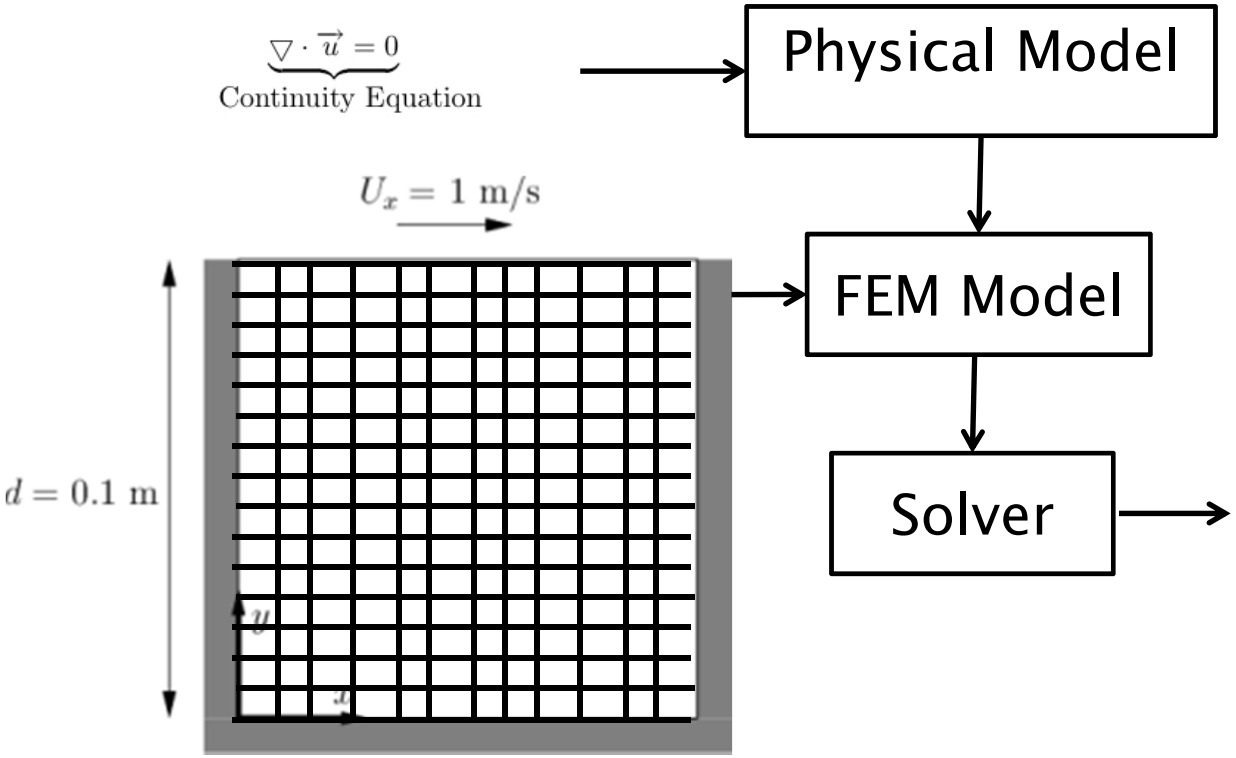
(1974)
978)
)
991)

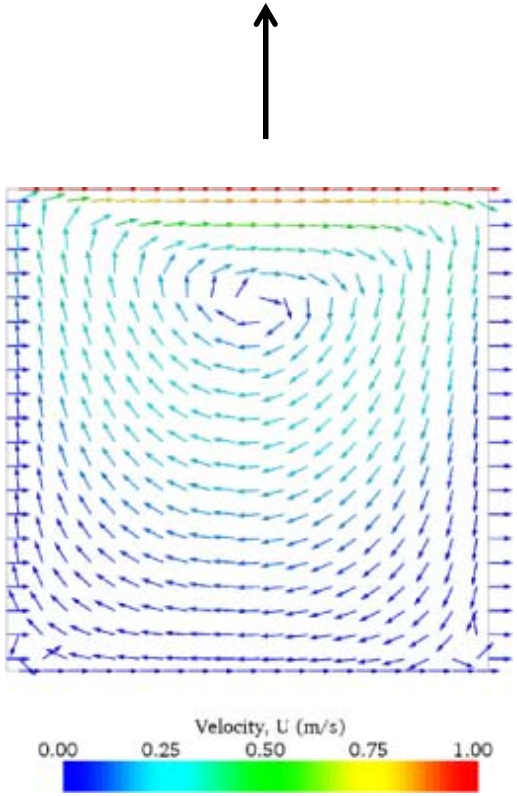
Lid-driven cavity flow

$$\rho\,(\underbrace{\frac{\partial \vec{u}}{\partial t} + \vec{u}\cdot\nabla\vec{u}}_{\text{Acceleration}}) = \underbrace{-\nabla p}_{\text{Pressure}} + \underbrace{\nu\,\triangle\,\vec{u}}_{\text{Viscosity}}$$

If there a vortex close to the lid?

$$\underbrace{\nabla\cdot\vec{u} = 0}_{\text{Continuity Equation}}$$





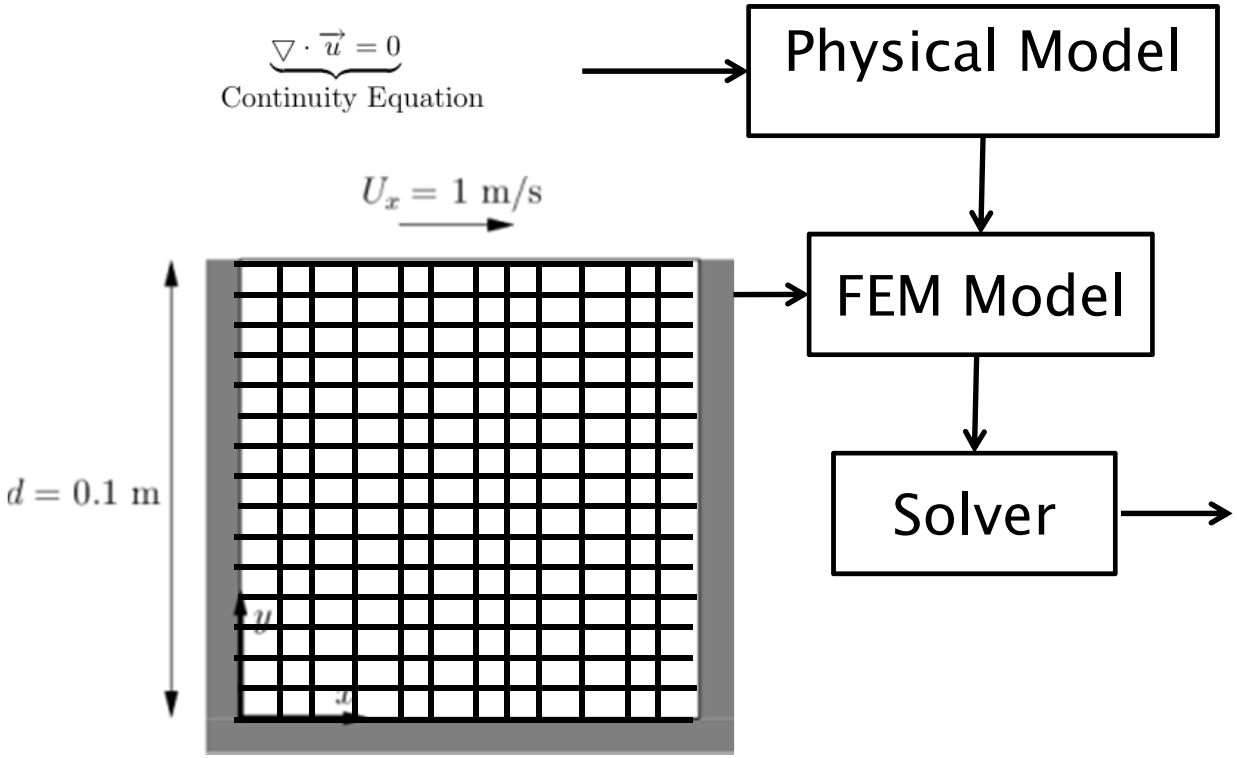
Lid-driven cavity flow

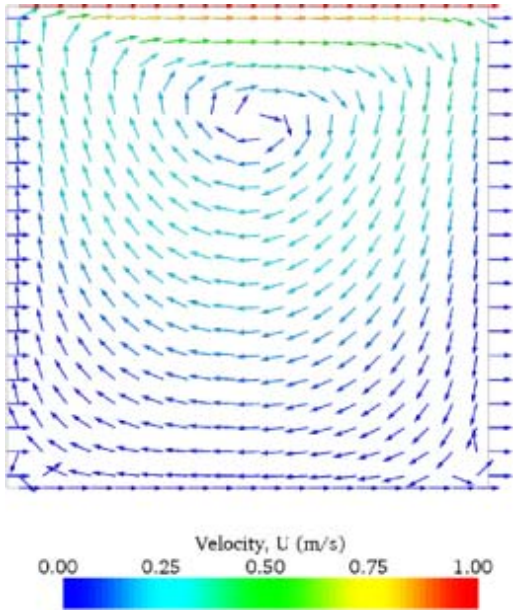
definition of a simulation

$$\rho\,(\underbrace{\frac{\partial \vec{u}}{\partial t} + \vec{u}\cdot\nabla\vec{u}}_{\text{Acceleration}}) = \underbrace{-\nabla p}_{\text{Pressure}} + \underbrace{\nu\triangle\vec{u}}_{\text{Viscosity}}$$

$$\underbrace{\nabla\cdot\vec{u}=0}_{\text{Continuity Equation}}$$

If there a vortex close to the lid?





Numerical Data Interpretation

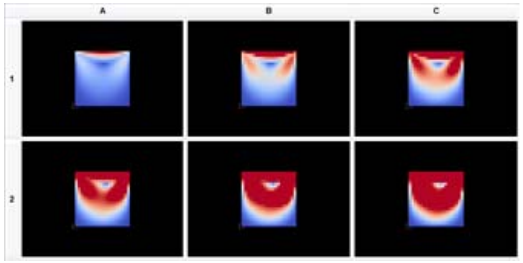
description of the simulation



Is translated into

Rules using references to anatomical, physical and data feature elements

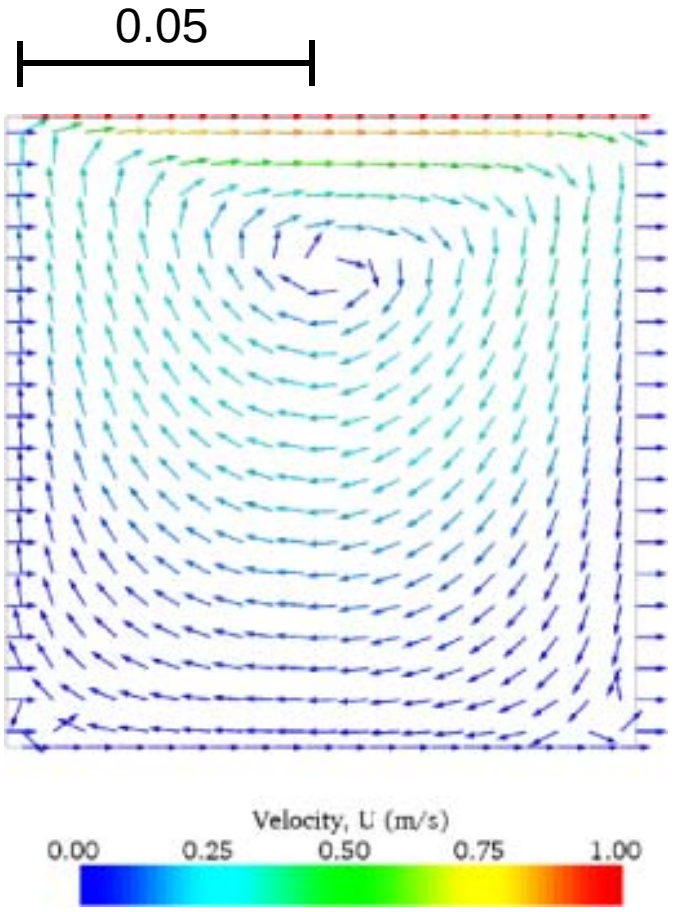
Multiple simulations



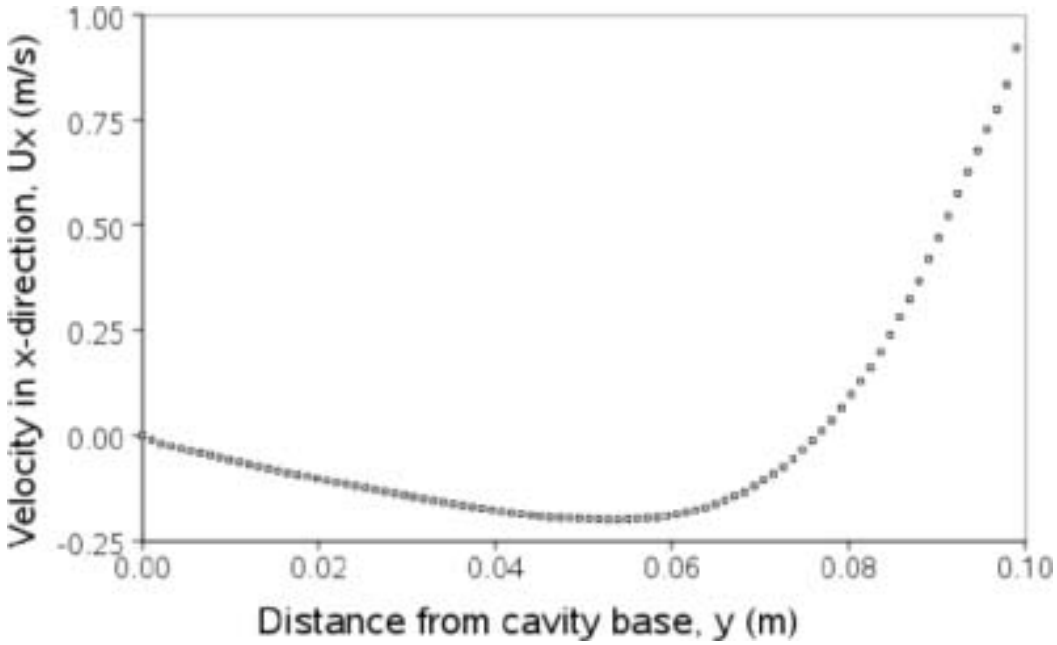
Feature extraction

Interpretation = rules applied over data at the symbolic level

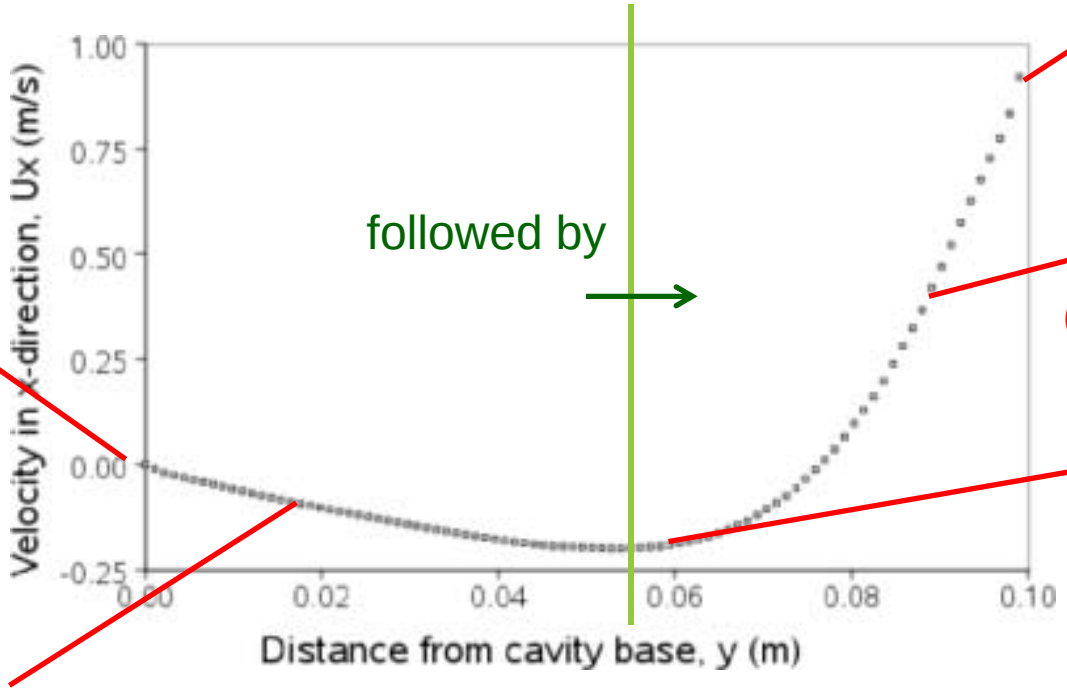
Data View



Data Selection

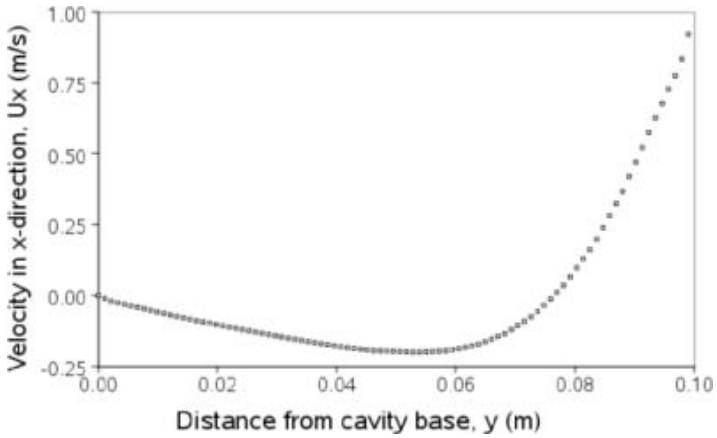


Feature Extraction



Based on the TEDDY ontology

Data Interpretation Statements



:DataView1 :hasDimensionY :VelocityX .
:DataView1 :hasDimensionX :DistanceFromTheCavityBase .
:DataView1 :x0 “0.0"^^xsd:double .
:DataView1 :y0 “0.0"^^xsd:double .
:DataView1 :hasMinimumX “-0.055"^^xsd:double .
:DataView1 :hasMinimumY “-0.20"^^xsd:double .
:DataView1 :hasFeature :PositiveSecondDerivative .
:DataView1 :hasBehaviour :BehaviourRegion1 .
:DataView1 :hasBehaviour :BehaviourRegion2 .
:BehaviourRegion1 :avgFirstDerivative “-3.63"^^xsd:double .
:BehaviourRegion1 :hasFeature EndRegion .
:BehaviourRegion1 :hasFeature :Decreases .
:BehaviourRegion1 :hasFeature :DecreasesSlowly .
:BehaviourRegion2 :avgFirstDerivative “33.35"^^xsd:double .
:BehaviourRegion2 :hasFeature EndRegion .
:BehaviourRegion2 :hasFeature :Increases .
:BehaviourRegion2 :hasFeature :IncreasesFast .
:BehaviourRegion1 :isFollowedBy :BehaviourRegion1 .
: LidSimulation :hasInterpretation :ValidVelocityBehaviour .

Data Analysis Rule

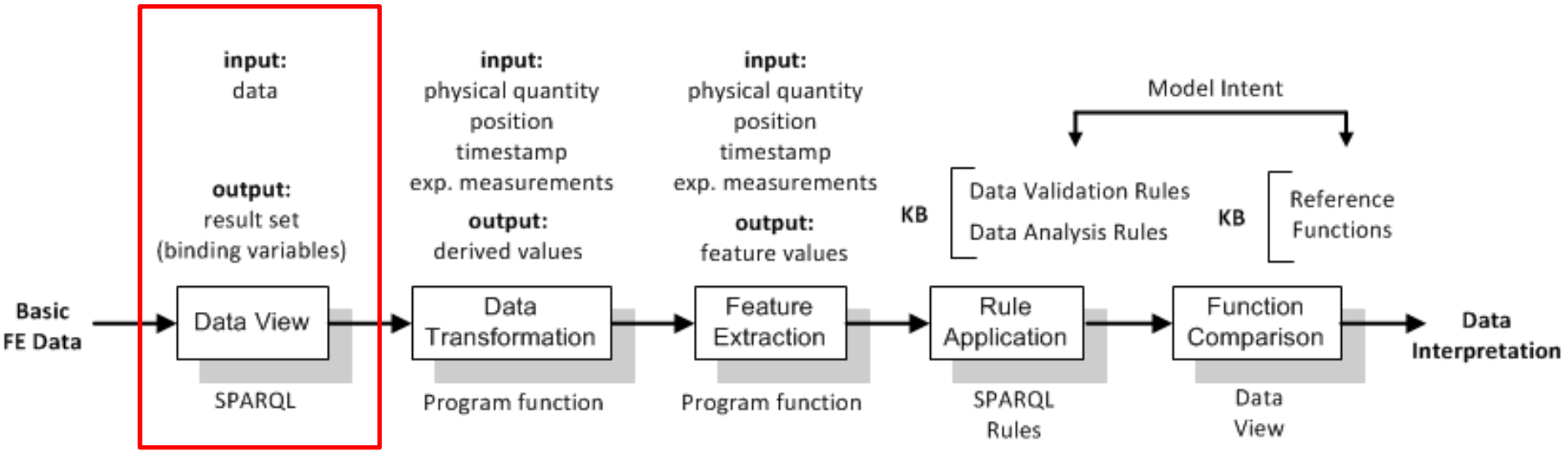
Data Analysis Rules

IF(minima(velocity) is negative **AND**
decreases very slowly(velocity) **AND**
increases very fast (velocity) **)**
VALID VELOCITY BEHAVIOUR

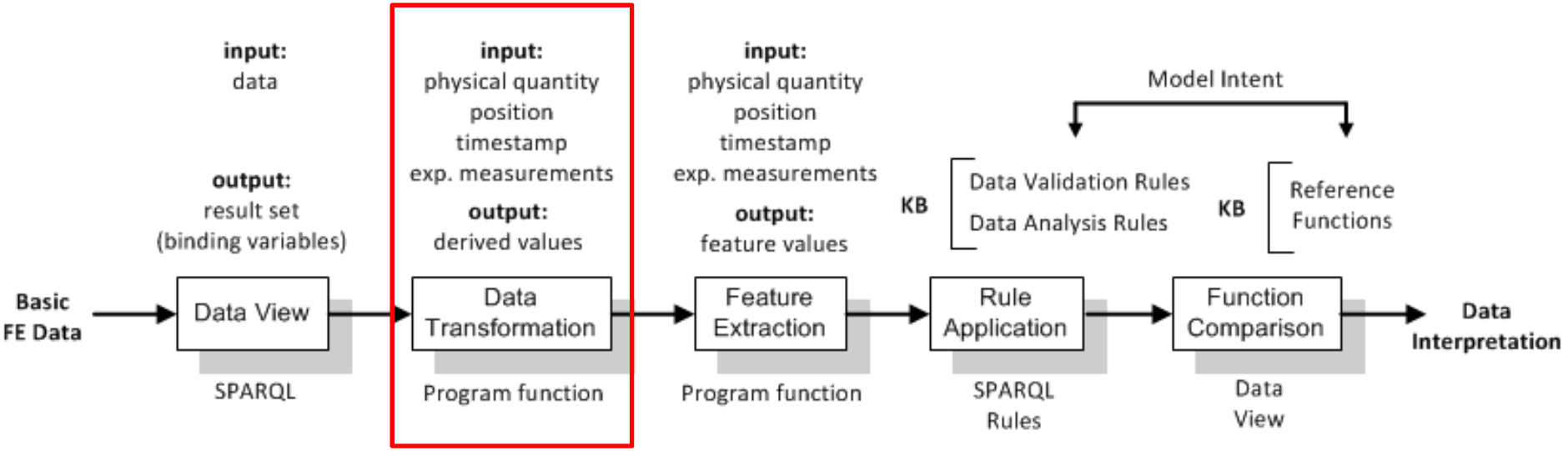
SPARQL Rule

```
CONSTRUCT
        { :LidSimulation sif: hasInterpretation :ValidVelocityBehaviour }
         WHERE {
            ?dataview rdf:type dao:DataView .
            ?dataview dao:hasFeature ?x .
           ...
        }
```

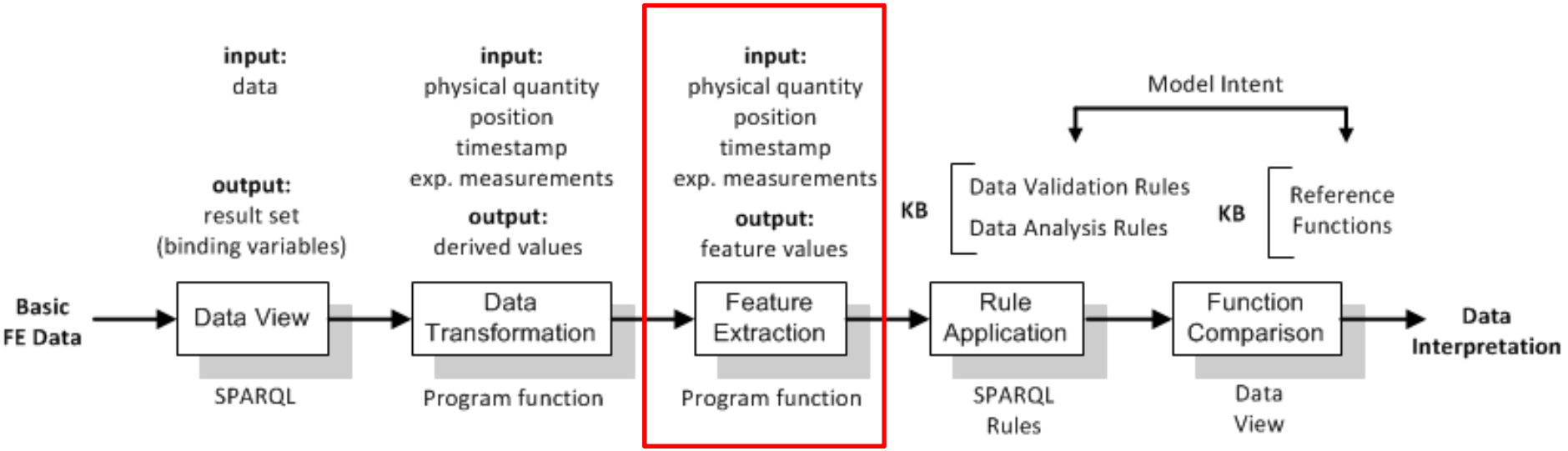
Data Analysis Workflow



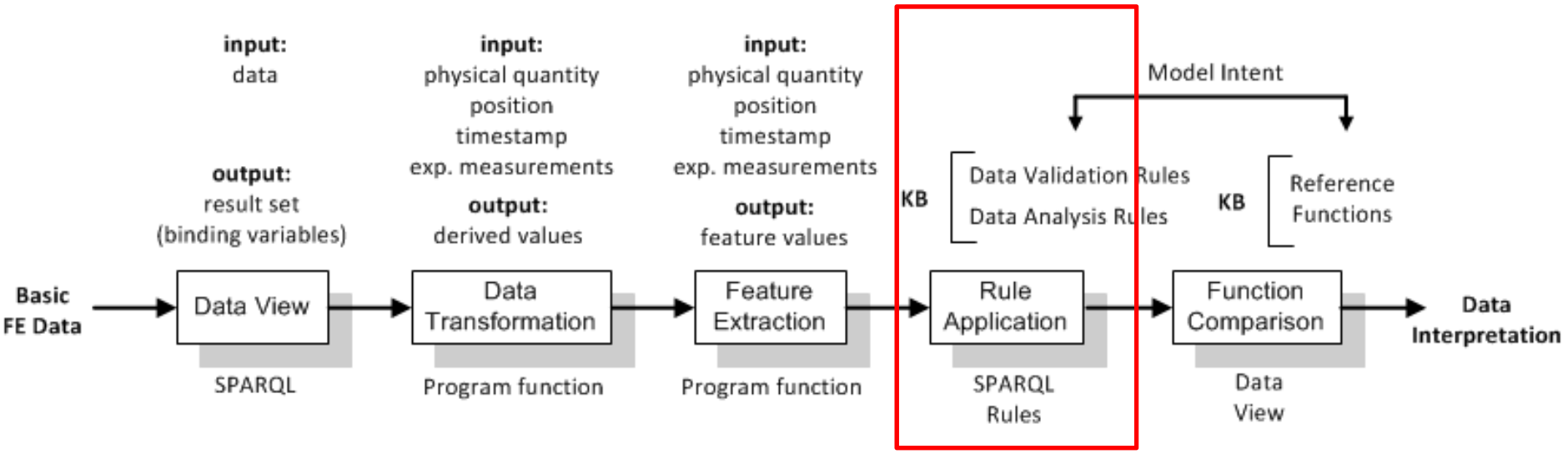
Data Analysis Workflow



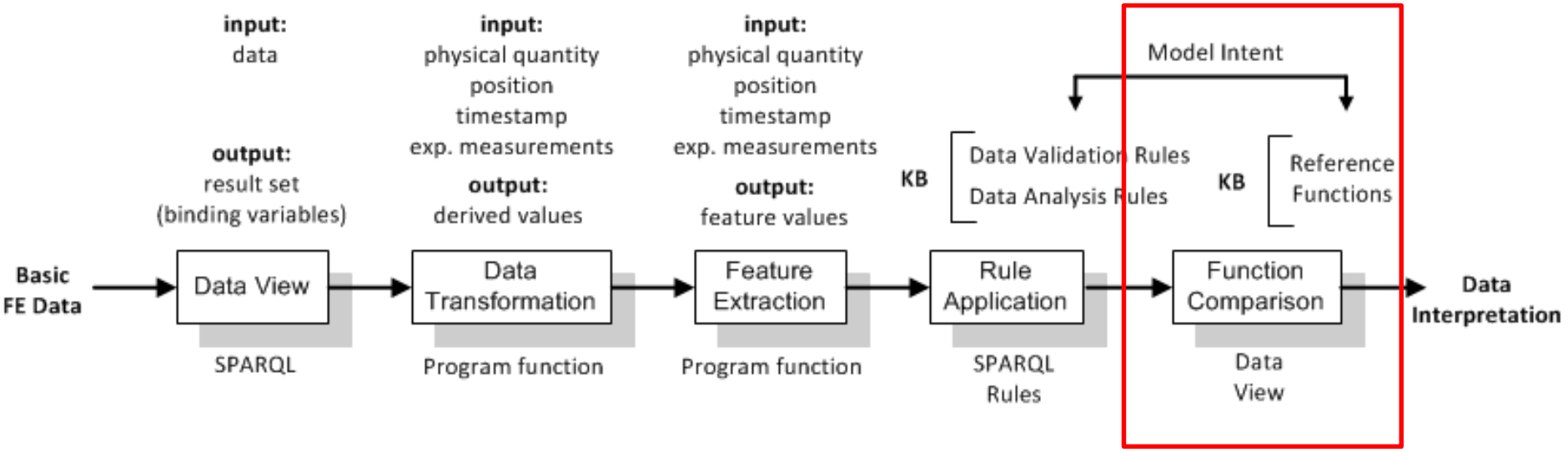
Data Analysis Workflow



Data Analysis Workflow

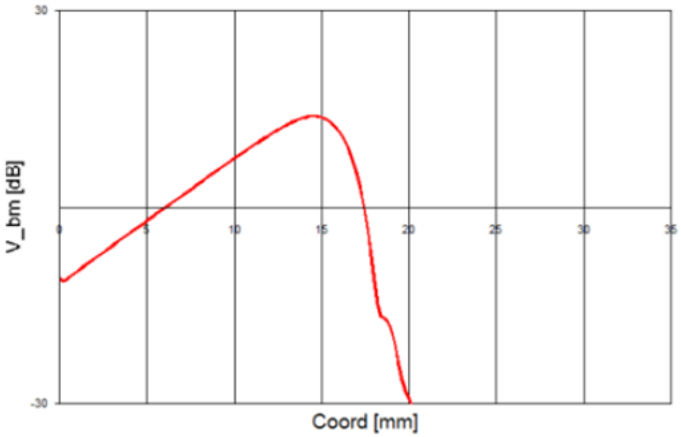


Data Analysis Workflow

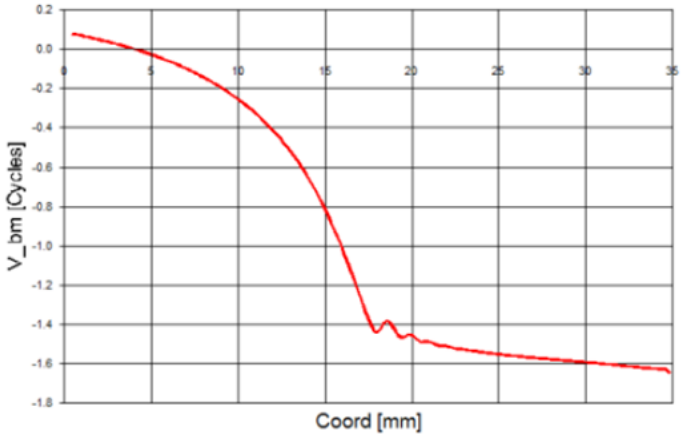


Output Data Views

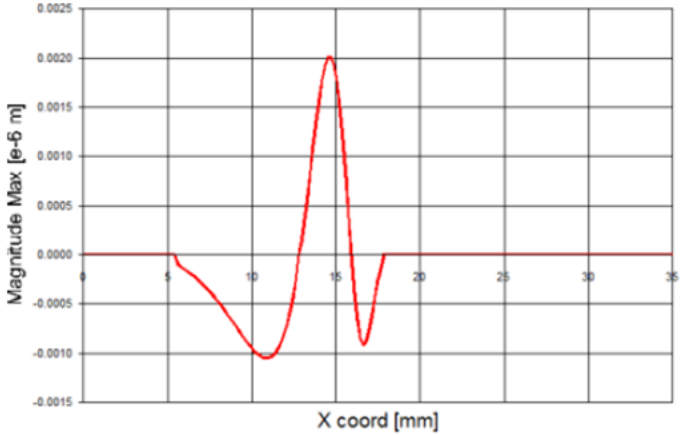
Modal BM Velocity Magnitude x Distance from BM basis



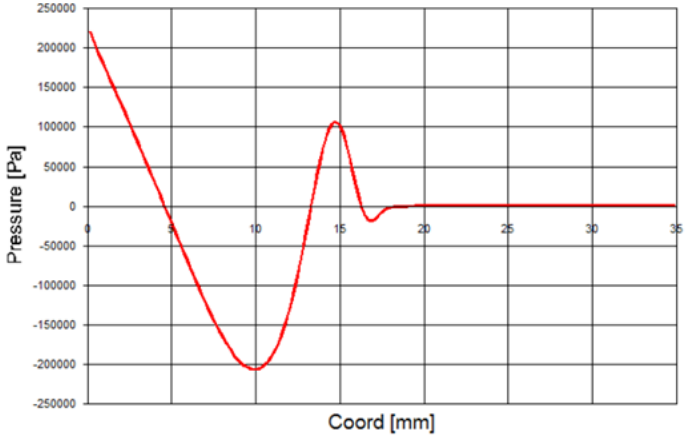
Modal BM Velocity Phase x Distance from BM basis



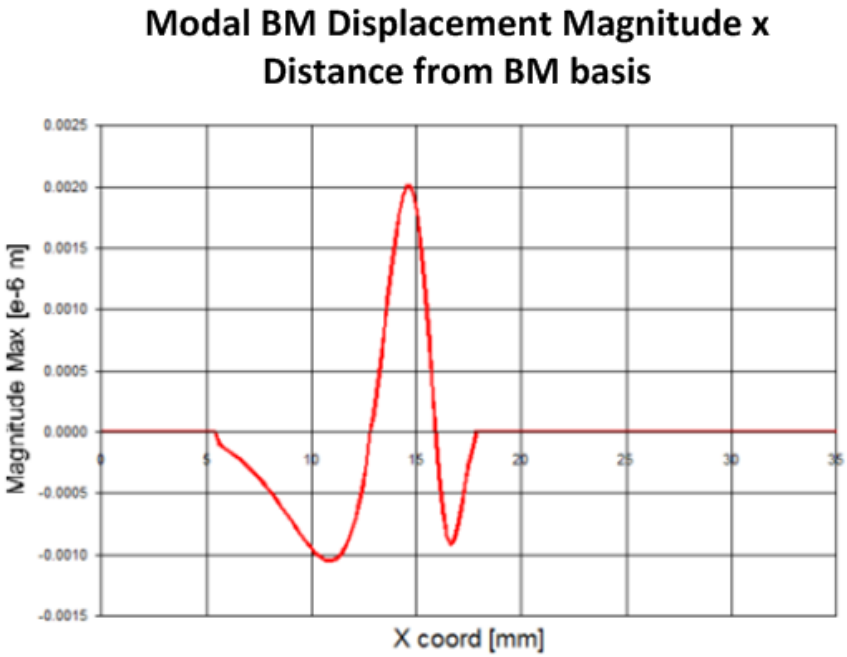
Modal BM Displacement Magnitude x Distance from BM basis



Real Part of Pressure x Distance from BM basis



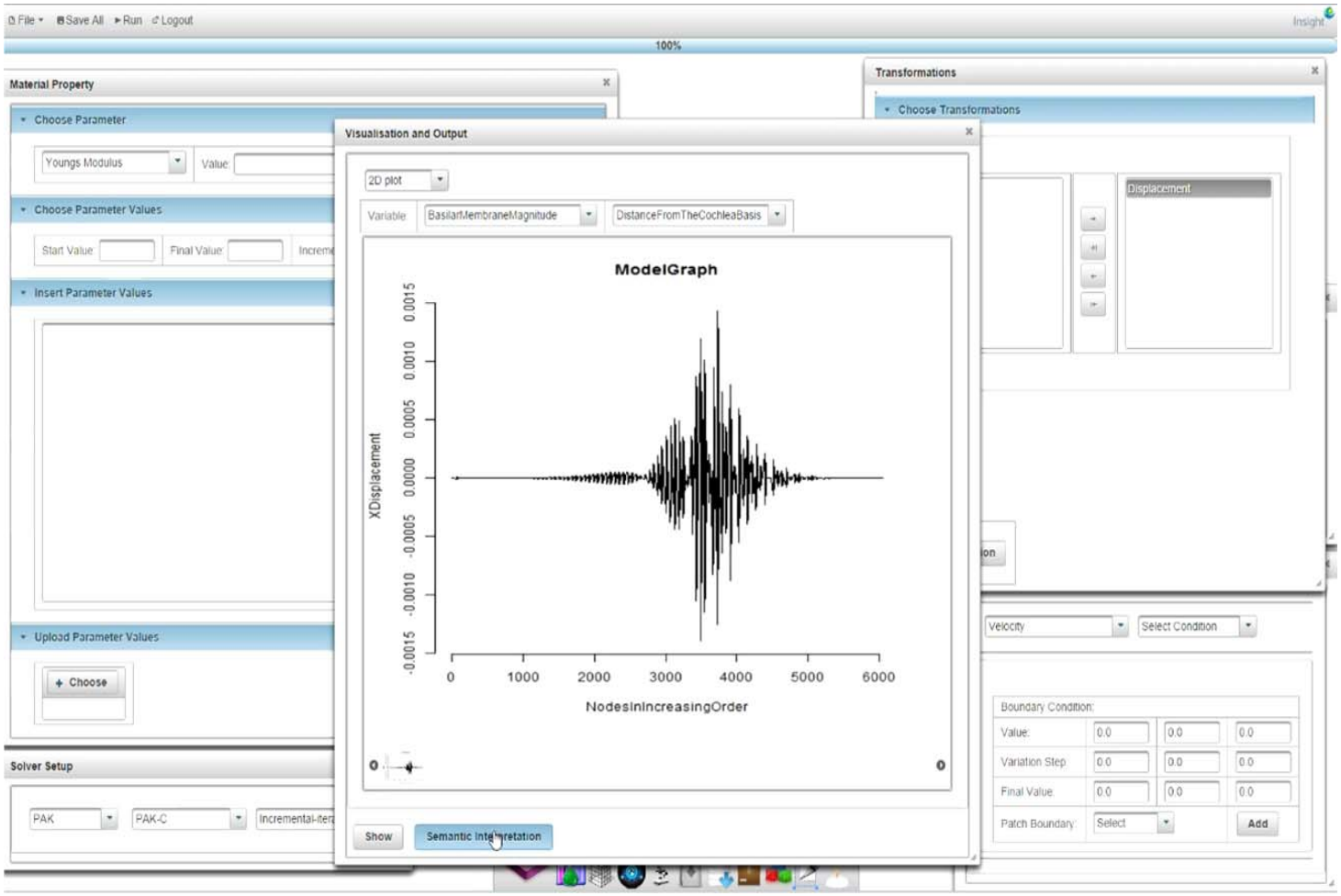
Feature Extraction



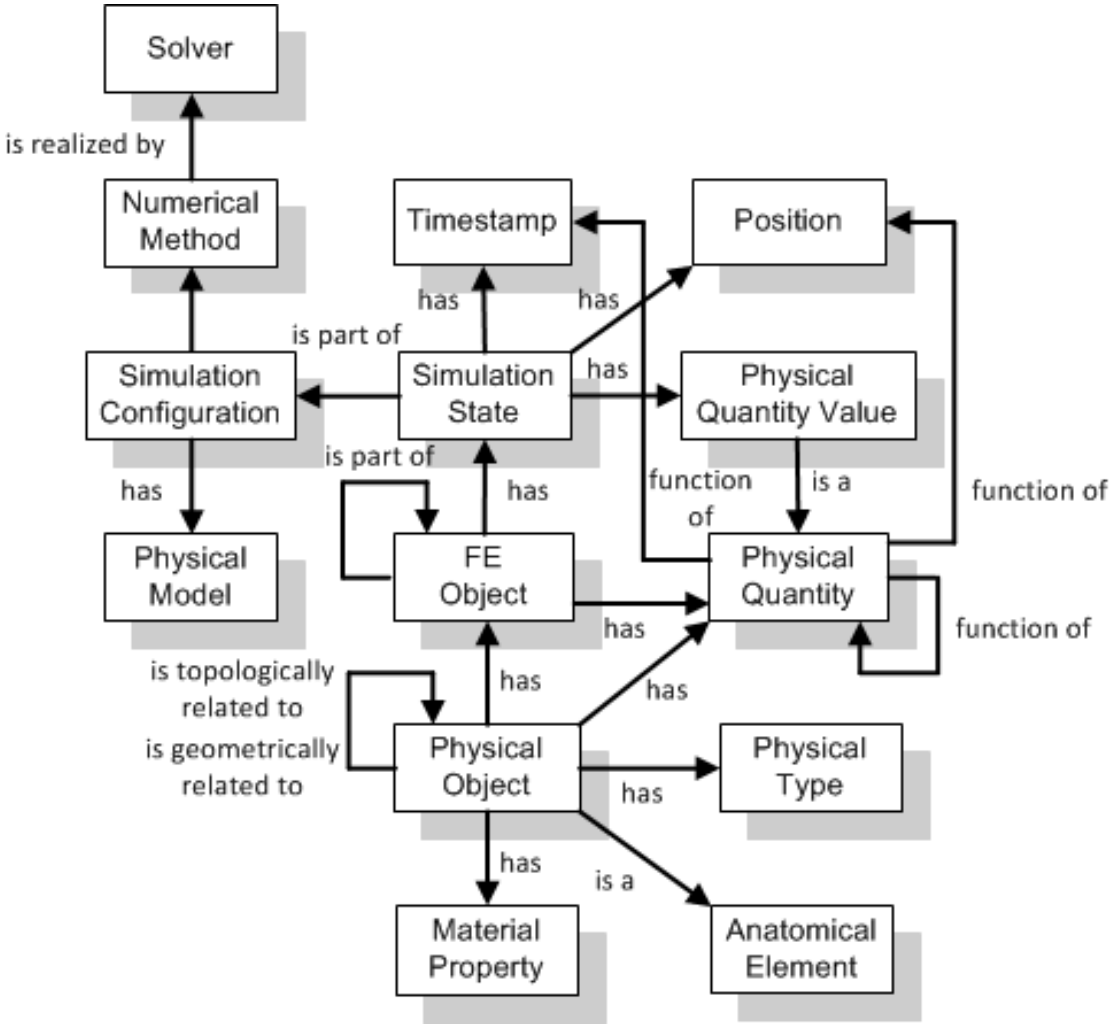
:DataView1 :hasDimensionY :BasilarMembraneMagnitude .
:DataView1 :hasDimensionX :DistanceFromTheCochleaBasis .
:DataView1 :hasFeature :isSingleWave .
:DataView1 :hasMaximumAmplitude “0.0031 "^^xsd:double.
:DataView1 :hasMaximumY “0.0020 e^-6 "^^xsd:double .
:DataView1 :hasMaximumX “14"^^xsd:double .
:DataView1 :hasMinimumY “-0.0011 e^-6 "^^xsd:double .
:DataView1 :hasMinimumX “17"^^xsd:double .

Demonstration (Video)

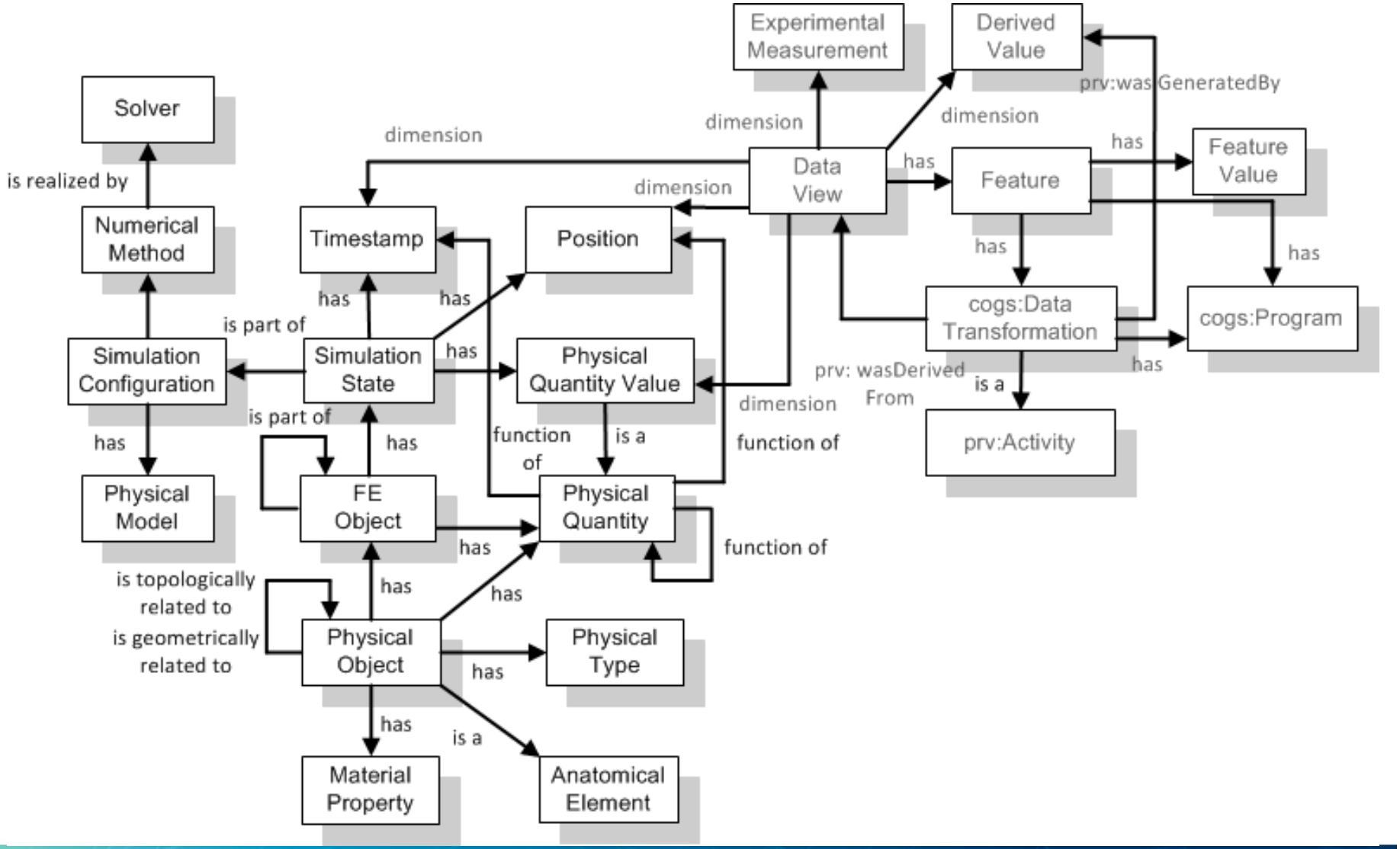
http://bit.ly/1rEZYh7



Conceptual Model Excerpt



Conceptual Model Excerpt



Take-away message

- Contemporary science demands new infrastructures to scale scientific discovery in a complex knowledge environment.
- Numerical data is everywhere, not only in FE simulations.
- In this work we started exploring how to represent and extract numerical data features to a conceptual level.

Future Directions

- **Better integration of the proposed representation and data analysis framework to the (TErminology for the Description of DYnamics) TEDDY conceptual model [EMBL-EBI].**

- **Use of the feature set and rules as a heuristic method to improve the simulation configuration space.**

SIFEM TEAM

- Andre Freitas
- Kartik Asooja
- Joao B. Jares
- Stefan Decker
- Ratnesh Sahay



SIFEM

http://sifem-project.eu

European Commission

Semantic Infostructure: interlinking an open source Finite Element tool and libraries with a model repository for the multi-scale modelling and 3D visualization of the inner-ear

Thank You !